

# Gouki Minegishi

Ph.D. Candidate · The University of Tokyo

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## RESEARCH INTERESTS

Mechanistic Interpretability

In-Context Learning

Large Language Models

Reasoning

Sparse Autoencoders

Neural Network Circuits

## EDUCATION

### Ph.D. in Computer Science

The University of Tokyo · Advisor: Prof. Yutaka Matsuo

Apr 2025 – Mar 2028

### M.S. in Computer Science

The University of Tokyo

Apr 2023 – Mar 2025

### B.S. in Engineering

The University of Tokyo

Apr 2019 – Mar 2023

## SELECTED PUBLICATIONS

### [1] Emergent Analogical Reasoning in Transformers

[Gouki Minegishi](#), Jingyuan Feng, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

*International Conference on Machine Learning (ICML)*, 2026 **Spotlight (top 2.2%)**

### [2] Steering at the Source: Style Modulation Heads for Robust Persona Control

Yoshihiro Izawa, [Gouki Minegishi](#), Koshi Eguchi, Sosuke Hosokawa, and Kenjiro Taura

*International Conference on Machine Learning (ICML)*, 2026

### [3] Understanding Emergent Misalignment via Feature Superposition Geometry

[Gouki Minegishi](#), Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026

### [4] Mechanism of Task-oriented Information Removal in In-context Learning

Hakaze Cho, Haolin Yang, [Gouki Minegishi](#), and Naoya Inoue

*International Conference on Learning Representations (ICLR)*, 2026

### [5] RL Squeezes, SFT Expands: A Comparative Study of Reasoning LLMs

Kohsei Matsutani, Shota Takashiro, [Gouki Minegishi](#), Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

*International Conference on Learning Representations (ICLR)*, 2026

### [6] LIT-RAGBench: Benchmarking Generator Capabilities of Large Language Models in Retrieval-Augmented Generation

Koki Itai, Shunichi Hasegawa, Yuta Yamamoto, [Gouki Minegishi](#), and Masaki Otsuki

*Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, 2026

### [7] In-Context Neurofeedback: Can Large Language Models Control Their Internal Representations through Privileged Access?

Koshiro Aoki, Ryota Takatsuki, [Gouki Minegishi](#), Yusuke Haruki, and Daisuke Kawahara

*Trustworthy AI for Good (AI4GOOD) Workshop @ ICML*, 2026 **Spotlight**

### [8] Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties

[Gouki Minegishi](#), Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

*Neural Information Processing Systems (NeurIPS)*, 2025

### [9] Beyond Induction Heads: In-Context Meta Learning Induces Multi-Phase Circuit Emergence

[Gouki Minegishi](#), Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo

[10] **Rethinking Evaluation of Sparse Autoencoders through the Representation of Polysemous Words**

Gouki Minegishi, Hiroki Furuta, Yusuke Iwasawa, and Yutaka Matsuo

International Conference on Learning Representations (ICLR), 2025

[11] **Interpreting Multi-Attribute Confounding through Numerical Attributes in Large Language Models**

Hirohane Takagi\*, Gouki Minegishi\*, Shota Kizawa, Issey Sukeda, Hitomi Yanaka (\* Equal Contribution)

International Joint Conference on NLP / Asia-Pacific Chapter of ACL (IJCNLP-AAACL), 2025

[12] **In-Context Meta Learning Induces Multi-Phase Circuit Emergence**

Gouki Minegishi, Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo

ICLR Workshop on Building Trust in Language Models and Applications, 2025 **Oral**

[13] **Bridging Lottery Ticket and Grokking: Understanding Grokking from Inner Structure of Networks**

Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo

Transactions on Machine Learning Research (TMLR)

[14] **Towards Empirical Interpretation of Internal Circuits and Properties in Grokked Transformers on Modular Polynomials**

Hiroki Furuta, Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo

Transactions on Machine Learning Research (TMLR)

[15] **ADOPT: Modified Adam Can Converge with Any  $\beta_2$  with the Optimal Rate**

Shohei Taniguchi, Keno Harada, Gouki Minegishi, et al.

Neural Information Processing Systems (NeurIPS), 2024

[16] **Interpreting Grokked Transformers in Complex Modular Arithmetic**

Hiroki Furuta, Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo

ICLR Workshop on Bridging the Gap Between Practice and Theory in Deep Learning, 2024 **Oral**

## RESEARCH & WORK EXPERIENCE

### Student Researcher

May 2026 – Oct 2026

Google DeepMind, Japan

### Research Scientist

Jun 2025 – Present

Third Intelligence, Inc.

Research on developing new forms of intelligence beyond conventional AI, aiming to bridge human-like reasoning, perception, and action.

### Chief AI Engineer

Nov 2024 – May 2025

Matsuo Institute, Inc.

Developed automated driving software, including 3D object detection with deep learning and integration of the World Model into automated driving.

### Research Intern

Aug 2024 – Oct 2024

Preferred Networks, Inc.

Research on in-context learning in foundation models, with knowledge conflicts.

## ACADEMIC HONORS

Best Paper Award, JSAI 2026 2026

Spotlight Presentation (top 2.2%), ICML 2026 2026

Spotlight Presentation, ICML 2026 AI4GOOD Workshop 2026

Young Researcher Encouragement Award, NLP 2025

Outstanding Presentation Award, JSAI 2024 2024

Oral Presentation, ICLR 2025 Workshop (Building Trust in LMs) 2025

Oral Presentation, ICLR 2024 Workshop (Bridging Practice and Theory) 2024

Selected for "BOOST NAIS" Scholarship Program for fostering advanced AI talents

## ACADEMIC SERVICE

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### Reviewer:

- **ICML** — 2026 (Silver Reviewer), 2025
- **NeurIPS** — 2026, 2025
- **ICLR** — 2026
- **AAAI** — 2026
- **TMLR**

**Organizer:** Mechanistic Interpretability Organized Session at JSAI 2025